

46. Primary and Secondary Hypertension · assets 1-9 / 18

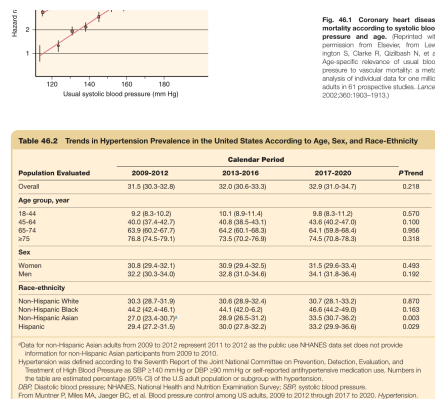


Fig 46 1

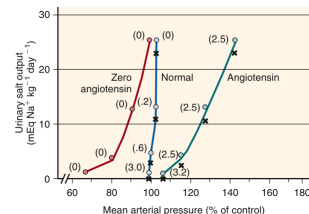


Fig. 4.62 Pressure-natriuresis curves in dogs at different levels of sodium intake [reflected in urinary salt output, y -axis]. In the normal state, massive fluctuations in sodium intake produce minor changes in blood pressure (BP). In states of high angiotensin II (Ang II) levels (ang II infusion in this model), BP increases significantly, even with modest increases in sodium intake. Conversely, in states where Ang II is absent or low (administration of captopril in this model), the increased sodium results in increased BP only at low BP levels; once BP becomes normal, the relationship is similar to that of normal kidneys. The curves represent the relative renal excretory concentration of Ang II (being the reference). From Guyton AC, Coleman TG, Young DB, et al. Salt balance and long-term blood pressure control. *Annu Rev Med*. 1980;31:15–27.)

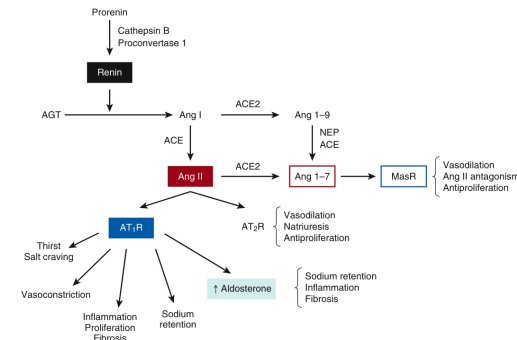


Fig. 46.3 Key elements of the renin-angiotensin-aldosterone system. ACE, Angiotensin-converting enzyme; ACE2, Angiotensin-converting enzyme type 2; Ang, angiotensin; AGT, angiotensinogen; AT1R, Ang II type 1 receptor; AT2R, Ang II type 2 receptor; MasR, Mas receptor; NEP, neutral endopeptidase.

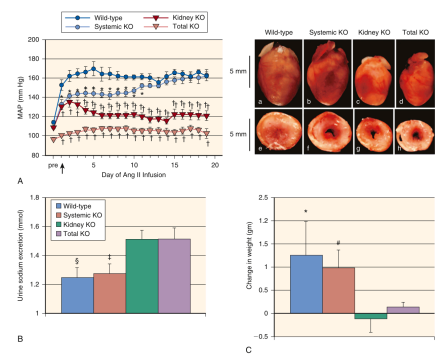


Fig. 46.4 Effects of angiotensin II (Ang II) infusion on blood pressure (A), urinary sodium excretion (B), body weight (C), and cardiac hypertrophy (photos) according to renal and extrarenal presence of Ang II type 1 receptor. See text for details. KO, Knockout; MAP, mean arterial pressure. (From Crowley SD, Gurley SB, Herrera MJ, et al. Ang II causes hypertension and cardiac hypertrophy through its receptors in the kidney. *Proc Natl Acad Sci U S A*. 2006;103:17985-17990. © 2006 National Academy of Sciences, U.S.A.)

Fig 46 4

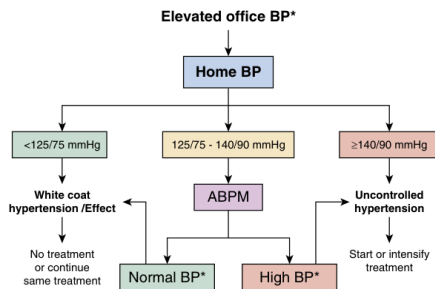


Fig. 46.7 Practical approach to the integration of home and 24-hour ambulatory blood pressure monitoring for the evaluation and management of hypertension.

Fig 46 7

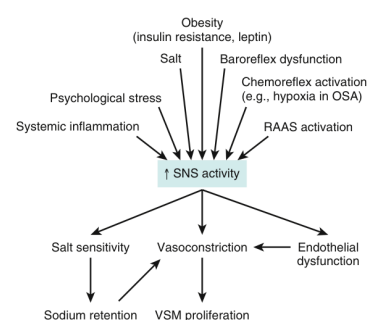


Fig. 46.5 Causes and consequences of sympathetic nervous system activation in the pathogenesis of hypertension. (OSA, Obstructive sleep apnea; RAAAS, renin-angiotensin-aldosterone system; SNS, sympathetic nervous system; VSM, vascular smooth muscle.)

Fig 46 5

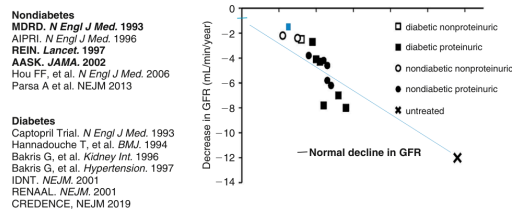


Fig. 46.8 Relationship between achieved blood pressure and decline in kidney function from primary renal endpoint trials. GFR, Glomerular filtration rate. (From Kalaitzidis R, Bakris GL. Update on reducing the development of diabetic kidney disease and cardiovascular death in diabetes. In Daugirdas JT, ed. *Handbook of Chronic Kidney Disease Management*. Philadelphia: Elsevier; 2018.)

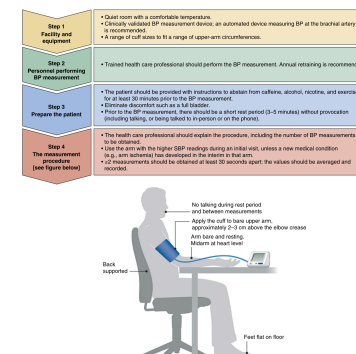


Figure 48-8 Steps for optimal blood pressure measurement. *From* Choung AK, Whorton PK, Muntner P, et al. International Consensus on Standardized Blood Pressure Measurement. *J Clin Hypertens*. 2010;12(1):1-11.

Fig 46 6

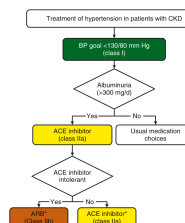


Fig. 46-9 Management of hypertension in patients with chronic kidney disease (CKD) according to American Heart Association/American College of Cardiology (AHA/ACC) guidelines.¹⁰ Colors indicate class of recommendation; green, class I [strong]; yellow, class II (moderate); orange, class III (weak). CKD stage 3 or higher = stage 1 or 2 with albuminuria ≥300 mg/dl or eGFR ≤30 mL/min/1.73 m². It should be noted that treating blood pressure in older patients to <140/90 mm Hg with an angiotensin receptor blocker or ACE inhibitor as first-line therapy is recommended for slowing progression of CKD by KDIGO guidelines. To further reduce CV risk, levels of LDL cholesterol need to be lowered in patients with CKD, especially with dIzemia in the subgroup with high albuminuria (has an additive reduction, in addition, to lower BP further compared with other calcium channel blockers, ACE, Angiotensin-converting enzyme, ARB), antidiabetic medications, statins, aspirin, and PPAR agonists. Abbreviations: ACC/AHA=American College of Cardiology/American Heart Association; AHA=American Heart Association; ACC=American College of Cardiology; JNC=Joint National Committee; NCCP=National Clinical Practice Guidelines; Hyperlipid=Hyperlipidemia; TIA=transient ischemic attack.

Fig 46 9

46. Primary and Secondary Hypertension · assets 10-18 / 18

Table 46.1 Categorization of Blood Pressure for Hypertension and Its Related Diagnoses in the United States^a

BP Category ^b	SBP		DBP
Normal	<120 mm Hg	and	<80 mm Hg
Elevated	120–129 mm Hg	and	<80 mm Hg
Hypertension			
• Stage 1	130–139 mm Hg	or	80–89 mm Hg
• Stage 2	140–159 mm Hg	or	90–99 mm Hg

^aIf the systolic and diastolic blood pressure levels fall into two different diagnostic categories, the higher category is used (e.g., 162/92 mm Hg is stage 3 hypertension; 134/72 mm Hg is stage 1).

*Definitions are from Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APHA/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: Executive Summary: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018;71:1269–1324. Different classification schemes are used outside the United States.

Table_46_1

Table 46.2 Trends in Hypertension Prevalence in the United States According to Age, Sex, and Race-Ethnicity

Population Evaluated	Calendar Period			P Trend
	2009-2012	2013-2016	2017-2020	
Overall	31.5 (30.3-32.8)	32.0 (30.6-33.3)	32.9 (31.0-34.7)	0.218
Age group, year				
18-44	9.2 (8.3-10.2)	10.1 (8.9-11.4)	9.8 (8.3-11.2)	0.570
45-64	40.0 (37.4-42.7)	40.0 (38.5-43.1)	43.6 (40.2-47.0)	0.100
65-74	63.9 (60.2-67.7)	64.2 (60.1-68.3)	64.1 (59.8-68.6)	0.916
≥75	76.8 (74.5-79.1)	73.5 (70.2-76.9)	74.5 (70.8-78.3)	0.358
Sex				
Women	30.8 (29.4-32.9)	30.9 (29.4-32.5)	31.5 (29.6-33.4)	0.493
Men	32.2 (30.3-34.0)	32.8 (31.0-34.6)	34.1 (31.8-36.4)	0.192
Race-ethnicity				
Non-Hispanic White	30.3 (28.7-31.9)	30.8 (28.9-32.4)	30.7 (28.1-33.2)	0.870
Non-Hispanic Black	44.2 (42.4-46.1)	44.1 (42.0-46.2)	46.6 (44.2-49.0)	0.163
Non-Hispanic Asian	29.7 (23.4-30.7)*	28.9 (26.5-31.2)	33.5 (30.7-36.2)	0.003
Hispanic	29.4 (27.2-31.5)	30.0 (27.8-32.2)	30.0 (27.8-32.6)	0.029

Data for non-Hispanic Asian adults from 2009 to 2012 represent 2011 to 2012 as the public use NHANES data set does not provide information for non-Hispanic Asian participants from 2009 to 2010. Hypertension was defined according to the Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure as SBP ≥ 140 mm Hg or DBP ≥ 90 mm Hg or self-reported antihypertensive medication use. Numbers in the table are estimated percentage (95% CI) of the U.S. adult population or subgroup with hypertension. DBP: Diastolic blood pressure; NHANES, National Health and Nutrition Examination Survey; SBP, systolic blood pressure. From Munter P, Miles MA, Jaeger BC, et al. Blood pressure control among US adults, 2009 to 2012 through 2017 to 2020. *Hypertension*.

Table_46_2

Table 46.4 Indications for 24-Hour Ambulatory and Home Blood Pressure Monitoring

Indication	Home BP Monitoring	ABPM	Comment
Identify white coat hypertension	++	+++	ABPM still the gold standard when patients have home BP values that are borderline (125-135/80-85 mm Hg)
Identify masked hypertension	++	+++	
Identify true-resistant hypertension	++	+++	
Evaluate borderline office BP values without target-organ damage	++	+++	
Evaluate nocturnal hypertension	—	+++	Home BP better for infrequent symptoms or paroxysms; ABPM better if frequent within a 24-h period
Evaluate labile hypertension	++	++	
Evaluate hypotensive symptoms	+++	++	
Evaluate autonomic dysfunction	+	++	Home BP useful to monitor orthostatic hypotension; ABPM useful to quantify supine hypertension and determine overall (average) BP levels
Clinical research (treatment, prognosis)	++	+++	

ABPM, Ambulatory blood pressure monitoring; BP, blood pressure.

Table_46_4

Table 46.7 Factors That May Cause False-Positive or False-Negative Results of the Aldosterone/Renin Ratio

Factor	False-Positives	False-Negatives
Aldosterone relatively high Renin relatively low	Potassium loading β -blockers; central α -adrenoceptor agonists; direct renin inhibitors; nonsteroidal antiinflammatory drugs; chronic kidney disease; sodium loading	Hypokalemia
Aldosterone relatively low Renin relatively high		Diuretics; ACE inhibitors, angiotensin receptor blockers; calcium channel blockers (dihydropyridines); acute sodium depletion

ACE, Angiotensin-converting enzyme.

Table_46_7

Table 46.8 Drugs Commonly Associated With Hypertension

Oral Contraceptives

- Nonsteroidal antiinflammatory drugs (NSAIDs, selective, and nonselective)
- Sympathomimetics: pseudoephedrine, phenylpropanolamine, phentermine, cocaine, amphetamines (prescription or illegal), Yohimbine (α -2 antagonist)
- Selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs)
- Monamine oxidase inhibitors (MAOIs)
- Cyclosporine and tacrolimus
- Erythropoietin and darbepoetin
- Corticosteroids, mineralocorticoids (fludrocortisone)
- Anti-VEGF antibodies (bevacizumab, ramucirumab) and certain tyrosine kinase inhibitors with anti-VEGF activity (e.g., sorafenib, sunitinib, vandetanib)
- Protease inhibitors (carilzomib)
- 11- β -hydroxysteroid dehydrogenase type 2 inhibitors:
 - Iclicore, pasonacozole
- Ethanol

From Peixoto AJ. Secondary hypertension. In: Gilbert SJ, Weiner DE, Bombardieri AS, Perazella MA, Rifkin DE, eds. *National Kidney Foundation's Primer on Kidney Diseases*. Philadelphia: Elsevier; 2023.

Table_46_8

Table 46.3 Clinical Clues to Guide the Investigation in Young Hypertensive Patients With a Potential Secondary Cause

[illegible]

Modified from Peleto AJ. Attending rounds: a young patient with a family history of hypertension. *Clin J Am Soc Nephrol*. 2014;9:2164–2172.

Table_46_3

Table 46.6 Initial Laboratory Evaluation of the Hypertensive Patient*

Test	Clinical Usefulness
Serum creatinine (and estimated glomerular filtration rate)	Assessment of renal function. Identifies parenchymal kidney disease as a possible secondary cause, as well as established TOD.
Serum potassium	Low potassium (of renal origin) suggests mineralocorticoid excess (primary or secondary), glucocorticoid excess, Liddle syndrome; high potassium with normal renal function suggests Gordon syndrome; low levels raise caution about the use of thiazides and loop diuretics; high levels preclude the use of ACE inhibitors, ARBs, renin inhibitors, and potassium-sparing diuretics.
Serum sodium	If high, suggests primary aldosteronism; if low, alerts to the need to avoid thiazide diuretics.
Serum bicarbonate	If high, suggests aldosterone excess (primary or secondary); if low with normal renal function, suggests Gordon syndrome (with high potassium) or primary hyperparathyroidism (with high calcium).
Serum calcium	If high, suggests primary hyperparathyroidism.
Serum glucose	Identifies prediabetes or diabetes; in the appropriate setting, suggests glucocorticoid excess, pheochromocytoma, or acromegaly.
Lipid profile	Identifies hyperlipidemia.
Hemoglobin, hematocrit	If high, in the absence of other hematologic abnormalities or underlying lung disease, suggests sleep apnea.
Urinalysis ^b	Proteinuria and hematuria identify a possible secondary cause (glomerulonephritis); proteinuria can also be a marker of TOD.
Electrocardiography	Identifies left ventricular hypertrophy, old myocardial infarction, or other ischemic changes; identifies conduction abnormalities that may preclude the use of β -blockers or nondihydropyridine CCBs.

*To investigate the presence of comorbid conditions, secondary causes, or established target-organ damage

*Some organizations recommend screening microalbuminuria as a more sensitive tool to identify early renal injury. The most recent guidelines do not recommend BUN measurement alone, whereas some recommend the measurement of uric acid (as a marker of cardiovascular risk) and thyroid-stimulating hormone (to more specifically screen for hypo- and hyperthyroidism).

ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; BUN, blood urea nitrogen; CCB, calcium channel blocker; TOD, target-organ damage.

Table_46_6

Table 46.9 Recommended Treatment Approaches to Hypertensive Emergencies

[illegible]

*Initiation, reinstatement, or adjustments of long-acting oral antihypertensive medications should take place during the first 6 to 12 hours. Oral drugs are chosen on the basis of conventional guidelines. Intravenous medications can be tapered on the basis of observed blood pressure levels after the addition of oral agents.

²Nitroprusside continues to be a recommended agent by most guidelines. Given its potential toxicity, it should be avoided as a first choice when other options are available.

*Diffuse microvascular injury is identified as high-grade retinopathy (hemorrhages, exudates, or papilledema), acute kidney injury, or microangiopathic hemolytic anemia or thrombocytopenia present alone or in combination.

Table_46_9